

First-Principles Simulations of Chemical Transformations in Nanoporous Materials and Industrial Catalysts

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Keywords

nanoporous materials, reaction, diffusion, adsorption, machine learning, molecular dynamics

Abstract

Nanoporous materials including zeolites, metal-organic frameworks, and covalent organic frameworks offer high tunability and surface area, making them ideally suited to address global challenges such as CO₂ capture and conversion, utilization of renewable feedstocks, and air purification. Molecular modeling is essential to enable atomic-scale design for optimal performance. Chemical transformations in these materials include not only catalytic reactions, but also local or global structural rearrangements and are strongly dependent on extreme operating conditions typical for industrial processes. The performance of industrial catalysts is governed by complex reaction networks and multiscale phenomena like diffusion and reactions, spanning a broad range of timescales and length scales. Recent advances at the intersection of quantum mechanics, statistical physics, and machine learning have significantly improved our ability to model complex chemical transformations in industrial catalysts and nanoporous materials. Herein, we review current modeling strategies and highlight future directions for predictive, multiscale simulations of nanoporous catalysts under realistic conditions.

PES: potential energy surface

QM: quantum mechanical

ML: machine learning

DFT: density functional theory

MLP: machine learning potential

1. INTRODUCTION

Chemical transformations encompass all processes where the chemical bonding nature changes. Nanoporous materials are vital for various applications such as catalysis, molecular separations, and energy storage (1–3). They are tractable due to their characteristic pore dimensions, high surface area, porosity, and diversity. Chemical transformations taking place in nanoporous materials encompass not only catalytic reactions, active site engineering by the intentional creation of defects, and rearrangements of active sites upon working conditions, but also processes involving structural transitions in which the chemical architecture of the framework changes (4–8).

Following chemical transformations from a purely experimental point of view with atomic precision is extremely difficult. Experimentally, one can follow a top-down approach, where one zooms into the material using spectroscopy or imaging tools with the aim to obtain images or spectroscopic signals with the best possible resolution in both time and length scales (9–12). However, so far, it is still not possible to make a “molecular movie” with the most advanced spectroscopic or imaging techniques and to follow on the fly the breaking and formation of chemical bonds with atomic-scale resolution (13–17).

Modeling follows a bottom-up approach, where one starts from the atomistic scale, builds realistic structures for the material from scratch, and evaluates the material’s behavior under operando conditions (18). Each molecular modeling exercise consists of various steps. First, one needs to build a realistic structure of the nanoporous material; next, one needs a methodology to describe the interaction between the atoms. This leads to the construction of the potential energy surface (PES), which is a highly dimensional function in terms of the internal degrees of freedom. The method to construct the PES largely determines the accessible length scales and timescales of the simulations. For simulating chemical transformations where bonds are broken and formed, one needs quantum mechanical (QM) methods, but these are also computationally very expensive. The next step in the modeling exercise consists of finding interesting paths on the PES, bringing us to the area of sampling and statistical physics. Finally, one can derive thermodynamic and kinetic properties, including diffusion coefficients, free energy differences, and reaction rates.

A major challenge in modeling nanoporous materials used in industrial applications relates to their high degree of complexity, where many reactions can take place simultaneously, the behavior is strongly dependent on the operating conditions, and a multitude of events take place spanning a wide range of length scales and timescales (19). A crystal particle used in a catalytic or separation process has typical dimensions of approximately 50 nm to several micrometers (20), and timescales vary from picoseconds for molecular vibrations to seconds/hours for activated processes (21). Even with today’s available very strong high-performance computers, it is not possible to describe these multiscale phenomena in a consistent way where the interatomic interactions are described at the QM level and where sampling covers a broad range of timescales (22). Recently, machine learning (ML) techniques have made their entrance in the field of modeling industrially relevant processes, which allow the evaluation of energies and forces in a more efficient way (23–26). Thanks to these advances, it is now possible to access length scales and timescales that were previously out of reach using brute-force QM methods such as density functional theory (DFT). However, the derivation of machine learning potentials (MLPs) for industrially complex catalytic systems remains a challenge as one needs to generate training data that are highly accurate, and also cover the relevant chemistry for the system at hand.

To faithfully model chemical transformations in nanoporous materials, it is essential to follow the material dynamically at the conditions where the material does the work (19). Within this review, we give the current state of the art on modeling techniques for simulating chemical transformations taking place in nanoporous materials and industrial catalysts. We showcase how

operating conditions greatly affect the nature of intermediates and generate broad transition state regions. The methods are illustrated by examples taken from complex catalytic and separation systems such as the conversion of CO_2 or CH_3OH to high-value chemicals, capture, and separations taking place in metal-organic frameworks (MOFs). Significant methodological advances have also been made in other areas of catalysis, such as reactions occurring on metal surfaces (26, 27). A prominent example is the Haber-Bosch process—an essential industrial method for converting nitrogen and hydrogen into ammonia under extreme conditions. To highlight how such advancements can inspire the modeling of catalytic conversions in nanoporous solids, selected aspects from these fields are presented.

The review is organized as follows. We start by introducing key concepts of nanoporous materials, their structure, their properties, and their reactivity in Section 2. Within Section 3, the various steps of a molecular modeling exercise are introduced, and we explain how theoretical quantities can be related to experimental observations. Within Section 4, we provide key examples of why simulating at operating conditions is crucial in any modeling strategy. Section 5 gives more insight into the complexity of catalytic cycles and methods to construct reaction networks. In Section 6, we zoom in on the timescale and length scale of various events taking place. We end by enumerating some summary points and future issues.

2. NANOPOROUS MATERIALS

According to the International Union of Pure and Applied Chemistry convention, porous solids are categorized according to the size of the pores: micropores with pore sizes below 2 nm, mesopores with pore sizes between 2 and 50 nm, and macropores having pore sizes of more than 50 nm (28–30). Nanoporous materials are colloquially defined as porous solids having pores, cavities, or channels of nanometer-sized dimensions. Within the field of nanoporous materials, porous materials are classified as nanoporous as long as they have pore dimensions smaller than 100 nm (31). There are various classes of nanoporous materials, as illustrated in **Figure 1a** (1).

In this review, examples are mainly taken from ordered materials, including zeolites, MOFs, and covalent organic frameworks (COFs) (1, 8). The long-range order in crystalline nanoporous materials simplifies computational modeling by reducing the description of micrometer-sized crystallites to periodic repetitions of the unit cell of the crystal lattice or potentially supercells to also allow for the encapsulation of defects. However, this approach does not account for the presence of external surfaces of nanoporous particles and neglects possible surface interactions between the nanoparticles themselves or other noncatalytic components of a pelletized industrial catalyst, such as supports, promoters, and binder materials (32). The lack of long-range order in porous materials like mesoporous silica, porous carbon, and amorphous MOFs precludes a complete atomistic characterization, and modeling these materials remains a challenging task (33). Nanoporous materials are often characterized by exceptionally high internal surface areas, easily exceeding $1,000 \text{ m}^2/\text{g}$, comparable to approximately the area of a soccer field. An important aspect is their high degree of tunability at the nanoscale with various pore sizes and architectures (**Figure 1b**) (34). **Figure 1c** schematically illustrates how the structure of nanoporous materials can be modulated with atomic-scale precision by introducing new linkers, multiple active sites, different functional groups, etc.

MOFs are extremely versatile to modulate by means of the exchange of linkers and/or inorganic nodes to pursue specific properties of the material. Virtually the whole periodic table of elements becomes a playground to build periodic materials from molecular fragments (35–37). The strength of MOFs relies on their ability to encapsulate atomically precise functions at regular positions in the framework, originating from the perfect lattice or from intentionally created defects. It is

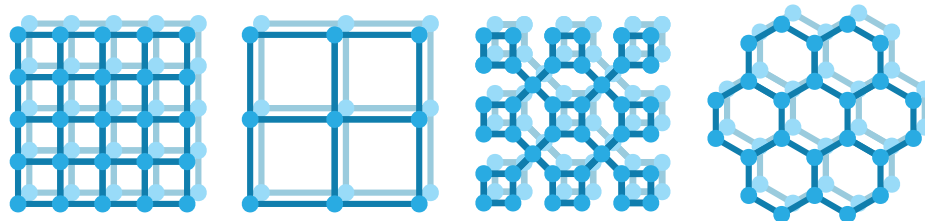
MOF: metal-organic framework

COF: covalent organic framework

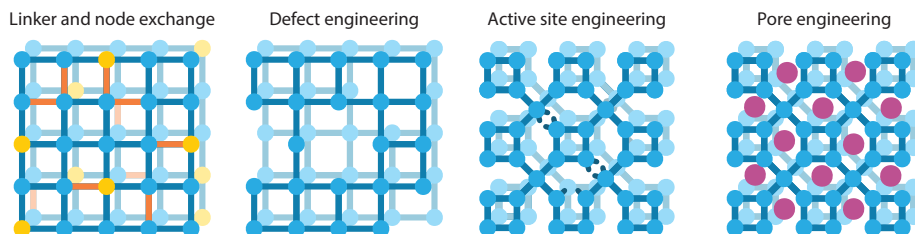
a Selected cases of nanoporous materials based on composition and pore size



b Pore size and architecture



c Design at the nanoscale



d Examples of functions and applications

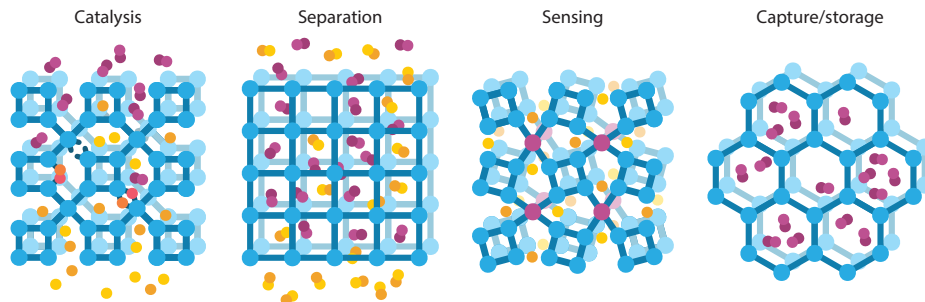


Figure 1

(a) Important classes of nanoporous materials, including zeolites, zeotypes, MOFs, COFs, mesoporous silicas, and porous (also called activated) carbons. Nanoporous materials are characterized by a wide variety of (b) pore sizes and architectures, which allows for precise control to (c) design new materials at the nanoscale for a (d) broad range of important applications, including catalysis, separation, sensing, and capture/storage of H_2 , CH_4 , CO_2 , volatile organic compounds, etc. Abbreviations: COFs, covalent organic frameworks; MOFs, metal-organic frameworks.

important to recognize that all realistic materials contain defects, ranging from point defects to extended mesopores (18, 21). Within the MOF field, this concept has been exploited to introduce additional functionalities into the framework, and it is now referred to as defect engineering (6).

Many of the unique properties of nanoporous materials are thanks to the near-molecular dimensions of their pores giving rise to nanoconfinement effects—a broad term that encompasses changes in thermodynamics, kinetics, mass transport, heat transfer, and the electronic behavior of guest molecules within the host material (38). Moreover, to process complex feedstocks, these materials offer the potential for active site engineering, to construct complex active sites where multiple ones cooperate (41–43). Furthermore, fluids in nanoconfinement (water, in particular) can have very different properties than in bulk, affecting the chemical reactivity of other guest molecules in the nanopores or even the properties of the nanoporous host material (44). At the scale of the pore, pore engineering can further enhance the properties of porous materials through precise control of the distribution, shape, size, and function of the pores. This can be achieved either by designing and selecting specific building blocks or through a wide range of postsynthetic modification methods (45–47). Promising concepts have been proposed to enforce thermodynamically difficult cascade reactions by maximally exploring shape selectivity and pore variations (40, 48). Shape selectivity can refer to reactant, transition, or product selectivity (39, 40).

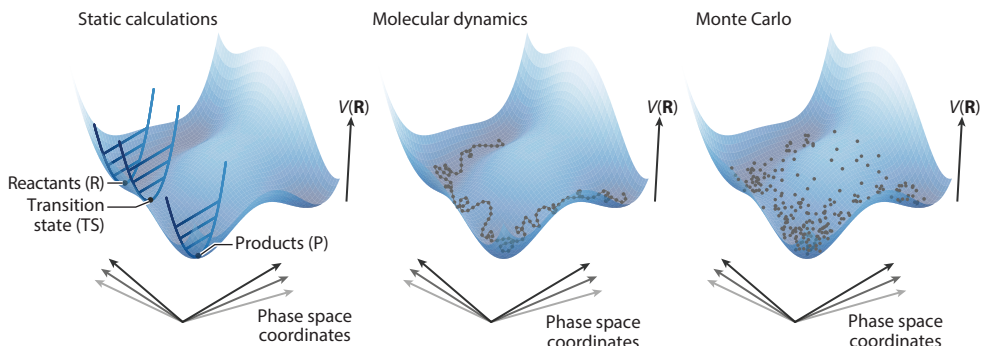
Nanoporous materials can have a multitude of functions, of which a selection is shown in **Figure 1d**. The function and state of the material critically depend on external conditions such as temperature, pressure, pH, and the presence of moisture. A first important function for many applications is a catalytic function where the materials are designed to include active sites to catalyze chemical reactions. Nanoporous materials show great promise as heterogeneous catalysts in thermo-, photo-, and electrocatalysis, such as overall water splitting to produce hydrogen and oxygen or catalytic CO₂ reduction. The inherent structure of MOFs as reticular frameworks of organic linkers and inorganic bricks (containing transition metals such as Ti, Zr, and Cu) results in a framework with open metal sites that can act as active sites, with the added option to functionalize the organic linkers to further enhance thermo-, photo-, and electrocatalytic activity. COFs, unlike MOFs, do not require (transition) metals but can be constructed from low-cost and abundant organic building blocks, making them excellent candidates as efficient and scalable photocatalysts. The inclusion of a separation function aims to discriminate between various molecules and is extremely important for industrial separations in an energy-efficient way (49–51). Nanoporous materials, in particular MOFs, can also be designed to include a sensing function, where the aim is to detect the presence or concentration of a specific molecule (e.g., volatile organic compounds) and to convert the material's response to a measurable signal (52–54). A capture function is related to the sensing function; however, here it is the intention to trap specific molecules for storage. Nanoporous materials are ideally suited for this purpose as they can include strong adsorption and binding sites in their framework to capture specific molecules. In this case, high uptake capacity and selectivity for the target molecule are important. Typical examples include CO₂ capture from flue gas using amine-functionalized MOFs or COFs, such as CALF-20 or COF-999 (34, 55). Other eye-catching applications include MOFs such as MOF-303 to collect water from the air (56, 57).

3. KEY STEPS OF MODELING CHEMICAL TRANSFORMATIONS IN NANOPOROUS MATERIALS VERSUS EXPERIMENT

To model chemical transformations in nanoporous materials, one can rely on a wide range of molecular simulation techniques to predict both thermodynamic properties—such as equilibrium constants and heats of adsorption—and kinetic properties—such as diffusivity and reaction

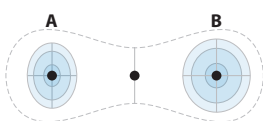
rates. Generally, a molecular modeling exercise consists of four main components, as illustrated in **Figure 2c**: (a) a realistic representation of the molecular structure of the material, (b) an adequate level of theory for evaluating the PES as a function of the atomic coordinates, (c) an efficient algorithm to generate PES samples that represent the phase space at the desired thermodynamic

a Potential energy surface exploration

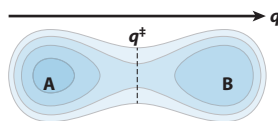


b Reaction rate calculation

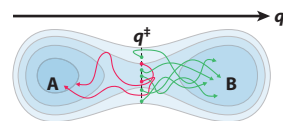
From harmonic approximation



From Landau free energies



Dynamically exact



c Sequence of a modeling exercise



d Functions and properties of nanoporous materials

Functions of nanoporous materials



Thermodynamic properties

Reaction (free) energy	
Equilibrium constant	
Adsorption (free) energy	
Adsorption isotherm	
Adsorption selectivity	
Heat capacity	

Kinetic properties

Reaction rate (of elementary step)	
Turnover frequency	
Arrival and residence time	
Diffusivity	
Surface permeability	
Thermal conductivity	

(Caption for Figure 2 appears on following page)

Figure 2 (Figure appears on preceding page)

(a) The PES can be explored using different sampling techniques: (*left*) static calculations after geometry optimizations, (*middle*) dynamic sampling using molecular dynamics to propagate a system forward in time, and (*right*) dynamic sampling using Metropolis Monte Carlo from the target probability distribution. (b) Reaction rates can be calculated using (*left*) static transition state theory, (*middle*) dynamic sampling of the free energy surface, or (*right*) dynamically exact Bennet-Chandler theory. (c) The sequence of a modeling exercise starts from a realistic atomic-level representation for the material to determine the multidimensional PES, which needs to be sampled (see also panel a) in the interesting regions at the desired thermodynamic conditions, after which the target properties can be derived. (d) Nonexhaustive overview of thermodynamic and kinetic properties that can be estimated from molecular modeling and their relevance for the various functions of nanoporous materials. Panel a adapted with permission from Reference 19 (CC BY-NC-ND 4.0). Panel c adapted from Reference 18 (CC BY 4.0). Abbreviations: FES, free energy surface; PES, potential energy surface; TST, transition state theory.

conditions (e.g., temperature, pressure), and (d) the reliable extraction of the properties of interest (18). In what follows, we discuss these aspects in more detail.

3.1. Molecular Representation of the Material

The first step in any molecular simulation protocol is building an adequate molecular representation of the material. Important choices include the use of isolated cluster models versus periodically extended crystals, inclusion of defects and other types of framework disorder, and accounting for framework flexibility (8). As highlighted in a recent perspective paper, this is a challenge in its own right, and interested readers are directed to that work for further guidance (18). Specifically for modeling chemical reactions occurring in nanopores, it is important to account for realistic loading of the pores with all species present in the reactant feed. Furthermore, the structure of realistic catalysts will dynamically evolve during operation; for example, in a typical methanol-to-hydrocarbons (MTH) experiment, one distinguishes various phases corresponding to the initiation, active catalysis, and deactivation phases (58). To account for these dynamic changes, it might be necessary to consider an ensemble of crystal particle configurations at various phases of the catalyst lifetime and take ensemble averages at various stages of the catalytic process (59). However, such ensemble averaging has to a lesser extent been performed in current modeling strategies within the field of catalysis.

3.2. Evaluation of the Potential Energy Surface

The second step involves computing the PES as a function of a given atomic configuration, which requires selecting an appropriate method to describe the interatomic interactions. When restricting ourselves to methods keeping atomic resolution, a distinction can be made between QM methods, classical force fields (FFs), and MLPs. Since chemical reactions involve the breaking and formation of bonds, FFs are generally not suited to describe these processes, as they assume a fixed chemical bond topology. Exceptions are reactive FFs, which are parameterized to allow chemical transformations (60). FFs are generally less accurate than QM methods—unless specifically fitted to reproduce experimental data. To define accuracy, one usually adopts the terminology “chemical accuracy,” which refers to methods that aim to obtain energies with an accuracy of 4 kJ/mol compared to exact benchmark values (61). For some applications, such as the calculation of adsorption isotherms, even this threshold may not be stringent enough, and very accurate QM methods are necessary (62). To model chemical transformations, QM methods are necessary in which the Schrödinger equation is solved for each nuclear configuration. In current modeling strategies, DFT has become the method of choice for QM descriptions of the system, thanks to its attractive trade-off between accuracy and speed. However, the accuracy of DFT is

MTH: methanol to hydrocarbons

FF: force field

MD: molecular dynamics

MC: Monte Carlo

CV: collective variable

TPS: transition path sampling

TIS: transition interface sampling

TST: transition state theory

largely dependent on the exchange-correlation functional and model for long-range dispersion interactions (63–72). While being fast among QM methods, DFT remains computationally relatively expensive, limiting the accessible length scales and timescales (22). Recently, MLPs, where the PES is described by a numerical potential derived from underlying QM training data, have become particularly popular (24, 25, 73–80). When trained correctly, MLPs offer the potential to evaluate energies and forces much more efficiently, while maintaining the accuracy of the underlying training data (25, 73–79, 81). The main challenge in developing MLPs suitable for chemical transformations in nanoporous materials lies in generating training data that accurately represent the relevant chemistry (78, 82, 83). Various strategies have been devised, including incremental learning, where an approximate MLP is used to systematically explore larger portions of the phase space, and cluster-based learning for disordered and defective systems, where training data are generated from smaller clusters cut from a periodic mesoscale structure (84, 85). Obviously, the final MLP will only be as accurate as the underlying training data, and sometimes it is necessary to go beyond DFT to train MLPs (86).

3.3. Sampling the Potential Energy Surface

Once the molecular structure and PES evaluation method are set, we need to choose how to sample the PES corresponding to the thermodynamic conditions (e.g., temperature, pressure) of interest. In general, one can identify two categories, i.e., static and dynamic sampling (see **Figure 2a**). Static sampling only considers a limited number of atomic configurations, e.g., (local) minima for reactant and product states and saddle points for transition states. When performing a normal mode analysis on the local configurations by diagonalizing the Hessian matrix, one can obtain thermodynamic and kinetic properties within the harmonic oscillator model. In dynamic sampling, the configurational state evolves either (a) deterministically by solving Newton's equations of motion on the PES, as in molecular dynamics (MD), or (b) stochastically by generating samples from a given probability distribution, as in Monte Carlo (MC) methods. MD and MC are complementary; for example, MD is ideally suited for diffusion or the vibration/deformation of a flexible framework, whereas MC is better suited for sampling adsorption (8, 19, 87, 88).

An important challenge is the simulation of rare events, i.e., events that spontaneously happen on a much larger timescale than typically accessible by MD/MC. Chemical reactions are prime examples of such rare events. There exist a plethora of techniques to enhance sampling of the low-probability regions. The first category relies on the definition of a collective variable (CV), which encodes the progress along the reaction under investigation. These methods improve sampling in regions of low probability that are critical to the reaction (89–93). A broad set of enhanced sampling methods exists (93–95). Definition of CVs is nontrivial for complex chemical transformations; in many cases this is done using chemical intuition. However, automatic methods have also recently been devised, which are also inspired by ML (96).

The second category involves path-based methods, including transition path sampling (TPS) and (replica exchange) transition interface sampling (TIS), which aim to sample an ensemble of dynamical trajectories undergoing the activated event without prior knowledge of transition states and the reaction coordinate (97–100). An ensemble of unbiased dynamical trajectories is created by systematically modifying a previous path in an MC fashion. From such simulations, one can extract dynamically correct reaction rates that go beyond the approximations of transition state theory (TST) (100, 101). So far, these methods have been less explored within heterogeneous catalysis, as many paths need to be evaluated, which is computationally very expensive when using QM methods. However, with MLPs, we foresee new avenues for path-based methods to obtain rates for complex chemical conversions (102).

Generally, the field has greatly evolved in the methods used to calculate rates of chemical reactions (**Figure 2b**). About two decades ago, reaction rates of chemical reactions within heterogeneous catalysis were mostly obtained from a static approach using information about reactants, transition states, and products, giving rise to a traditional free energy diagram with a discrete number of points (103, 104). With advances in first-principles MD techniques and stronger computational power, enhanced sampling strategies became more popular. Currently, a plethora of successful cases is available, showing how these methods give new insights into catalytic reactions at operating conditions. In most cases, one obtains the free energy profile (FEP) along with well-chosen CVs at the conditions of interest, after which reaction rates are derived in the limit of TST, accounting for broad transition and reactant regions, but still neglecting barrier recrossings (19, 105–111). With path-based methods, one can obtain more accurate rates (102). Recently, dynamically corrected proton hopping rates in zeolites were obtained using the Bennett-Chandler approach to reaction rate theory, explicitly accounting for barrier recrossings using the reactive flux formalism (101, 111–113). This approach relies on the calculation of many paths and was only possible thanks to the derivation of a reactive MLP (81).

FEP: free energy profile

GCMC: grand canonical Monte Carlo

TMMC: transition matrix Monte Carlo

3.4. Extraction of the Properties of Interest

In the last part of the molecular modeling exercise, properties of interest for the application at hand can be derived. **Figure 2d** gives an overview of common properties relevant for the topic of this review. Depending on the property and sampling scheme, this step can be as trivial as computing an ensemble average or it can require application of more advanced methodologies. For example, if grand canonical Monte Carlo (GCMC) simulations are done at varying chemical potentials, constructing the adsorption isotherm boils down to computing the average number of adsorbates at each chemical potential. However, as illustrated in the previous subsection, obtaining chemical reaction rates accounting for the true dynamics necessitates the combination of various steps, a proper sampling protocol, and its combination with methods to obtain the FEP and afterwards reaction rate theories. As the simulation conditions become more realistic, this last step of the molecular modeling exercise can become rather complex. A recent example illustrates this nicely where the adsorption isotherm for a flexible water-loaded MOF was derived using MD-based transition matrix Monte Carlo (TMMC) (62).

3.5. All Parts of the Molecular Exercise Are Connected

Finally, it is important to recognize that all steps of the molecular modeling exercise are connected, and choices at each step influence each other. This influence propagates not only forward along the modeling exercise but also backward; i.e., the nature of the property to be computed impacts the sampling scheme and required PES level of theory. For example, applying TST with a simple minimum-maximum approximation to the (free) energy barrier does not account for the dynamic freedom of the thermodynamic (meta)stable reactant state and will not result in reliable estimates of reaction rates. Therefore, one needs dynamic sampling, e.g., using enhanced sampling MD. Another example relates to the accurate prediction of adsorption isotherms for challenging systems. For MOF-303, a material used for water harvesting, accurate adsorption isotherms in agreement with the experiment could only be obtained using both an accurate level of theory (achieved by means of MLPs) and a proper account for framework flexibility (achieved by means of TMMC) (62). The methodology to generate the training data for adsorption simulations is distinctively different than that for regular MD simulations, as low-probability samples also need to be included in the training data to efficiently reject such configurations in the MC approach (114). Thus, depending on the property of interest, other choices need to be made in deriving the MLP, the sampling procedure, etc. (62).

4. THE IMPORTANCE OF OPERATING CONDITIONS FOR MODELING CHEMICAL TRANSFORMATIONS AND THE COMPLEXITY OF INDUSTRIAL CATALYSTS

Catalytic reactions taking place in industrial settings are mostly taking place under extreme conditions of temperatures and pressures. Typical examples are the Haber-Bosch process to industrially synthesize ammonia from nitrogen and hydrogen, typically operating under pressures from 20 to 25 MPa and temperatures between 400°C and 500°C or catalytic cracking over zeolites, operating at temperatures higher than 500°C (115, 116). Following the dynamics of heterogeneous catalysts under such extreme conditions is highly challenging from both an experimental and theoretical point of view. Looking at it from a holistic perspective, it is even surprising that materials remain stable under these conditions. The materials should maintain their structure and porosity when reactants and products travel to and from the active site. Moreover, under such operating conditions, catalysts do not exist in their most thermodynamically stable form and are therefore not structurally or chemically inert. Recently, Alexandrova & Christopher (117) coined the concept of catalysis taking place at phase boundaries, where a catalyst starts from a stable phase but must reach a region of instability that is catalytically useful. The challenge, then, is to find the appropriate conditions to bring the catalyst to the right phase boundary for the catalytic process under investigation. We expect more investigations along this direction in the future. In what follows, we discuss two studies showing the importance of operating conditions for industrial processes and also highlighting various aspects of the molecular modeling exercise.

4.1. Reactive Species at Operating Conditions

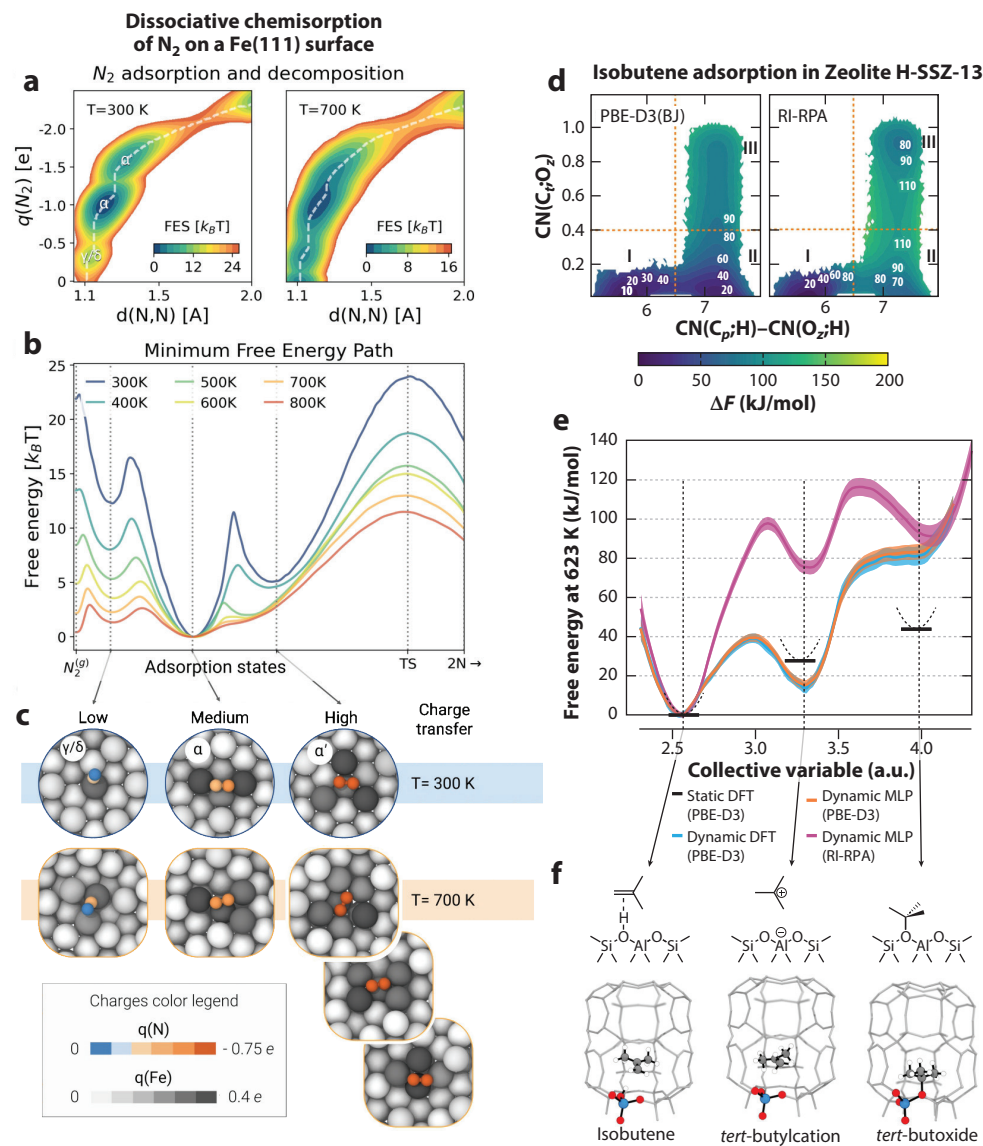
To illustrate the importance of operating conditions on the stability of reactive intermediates, an illustrative case is the nature of adsorbed species during alkene cracking taking place at very high temperatures, > 500°C (116). Such species are very difficult to track experimentally with in situ and operando techniques, as they can be very reactive and quickly decay into other intermediates. When an alkene adsorbs on a Brønsted acidic zeolite, four possible adsorbed states can be identified, corresponding to a physisorbed van der Waals complex, a physisorbed π -complex, a carbenium ion when the double bond is protonated by the Brønsted acid site, and, finally, a framework-bound alkoxide species if a covalent C–O bond is formed between the hydrocarbon and lattice (**Figure 3f**). The relative probability of finding each of these species depends on a series of factors, including the degree of branching of the hydrocarbon, the zeolite confinement, and the operating conditions (108, 118, 119).

To show how the various parts of the molecular modeling exercises are crucial to accurately predict the prevailing intermediates at operating conditions, we showcase the stability of adsorbed isobutene in H-SSZ-13. **Figure 3d–f** gives a summary of the main results. We have recently determined the FEP for the conversion of isobutene into *tert*-butylation and surface *tert*-butoxide with chemical accuracy, fully accounting for operating conditions (86). To reach such a degree of accuracy, we need both precise electronic energies and a correct description of entropy at the high temperatures at which conversions involving isobutene take place in industrial settings (~623 K).

With static calculations, the relative position of the various intermediates is not correctly predicted, as the dynamic movement of the mobile intermediates is not properly captured in the harmonic oscillator approximation (**Figure 3e**). More mobile physisorbed species will be entropically stabilized at higher temperatures, whereas chemically immobilized species like surface alkoxides will be entropically disfavored. To account for these effects, dynamics simulations performed in the literature showed a distinct influence on the FEP by correctly accounting for the mobility of the various species (108, 119, 120). However, a serious limitation of first-principles

MD simulations is their computational expense. To perform a 100-ps-long MD simulation using a typical time step of 0.5 fs, 200,000 PES evaluations are necessary. Even when using a low-cost DFT functional, extensive sampling to explore various zeolite topologies, operating conditions, and other parameters is not feasible as it would require millions of energy evaluations. For carbenium ions, moreover, DFT calculations themselves are known to not be accurate enough, due to charge delocalization error, and more expensive wave function–based methods are needed (64, 121). Performing brute-force MD simulations at these higher levels of theory is beyond reach even on the most powerful high-performance computers available today.

To address these issues, Bocus et al. (86) harnessed the power of MLPs, where they developed a transfer learning scheme to derive a highly accurate MLP approaching chemical accuracy for the



(Caption for Figure 3 appears on following page)

Figure 3 (Figure appears on preceding page)

(Left column) Dynamics, adsorption, and bond breaking of N_2 interacting with the Fe(111) surface. (a) Free energy as a function of N–N distance and the partial charge. Local minima represent the metastable states, and white dashed lines denote the minimum free energy pathways in this plane. (b) Minimum free energy pathways from the gas phase to the adsorption states to the dissociated state. (c) Snapshots of representative geometries of the adsorption states based on the amount of charge transferred for $T = 300$ K and $T = 700$ K. Atoms are colored according to charges predicted by the neural network model, with two different color scales for the nitrogen and iron atoms. Panels *a*, *b*, and *c* reproduced with permission from Reference 26 (CC BY-NC-ND 4.0). (Right column) Conversion of isobutene into *tert*-butylation and surface *tert*-butoxide in H-SSZ-13. (d) 2D FES from enhanced sampling using MLPs trained to PBE-D3(BJ) (left) and RI-RPA (right) data. (e) Corresponding projected 1D profiles to compare DFT-trained MLP (orange) with RI-RPA-trained MLP (purple), with static DFT (black) and dynamic DFT (blue). (f) The three investigated intermediates formed upon isobutene adsorption in Brønsted acidic zeolites. C is depicted in gray, Al in blue, O in red, and H in white; the rest of the framework is shown in gray sticks for the sake of clarity. Panels *d*, *e*, and *f* adapted with permission from Reference 86 (CC BY-NC-ND 4.0). Abbreviations: DFT, density functional theory; FES, free energy surface; MLPs, machine learning potentials; RI-RPA, resolution of identity random phase approximation; TS, transition state.

isobutene conversion in H-SSZ-13. First, an MLP was trained at a low DFT (PBE-D3) level of theory, using an active learning strategy in which a provisional MLP is used to explore (potentially new) regions on the PES, which are then evaluated and added to the training data for the following MLP generation. For this specific reaction, approximately 3,700 training structures needed to be labeled with PBE-D3 energies and forces to obtain an MLP capable of generating a free energy surface (FES) with the same accuracy as the reference DFT (Figure 3e). To improve the level of theory, 200 structures were subsampled from the training set and evaluated at the highly accurate resolution of identity random phase approximation (RI-RPA). Additional readout layers were added to the MLP and trained on the energy difference between the DFT level of theory (PBE-D3) and the target level of theory (RI-RPA). The results show that the FES changes enormously when moving from PBE-D3 to RI-RPA (Figure 3d,e). The *tert*-butylation, in particular, goes from being ~ 20 kJ/mol higher in energy than isobutene to ~ 80 kJ/mol. Hence, while the DFT level of theory predicts quite a significant concentration of *tert*-butylation in H-SSZ-13 at operating conditions, the more accurate calculations show that these intermediates would only be present at very low concentrations and lifetimes. This case study clearly illustrates the importance of operating conditions to determine the nature of elusive intermediates and nicely illustrates recent advances in the field at the intersection of QM-based methods, advanced sampling, and ML.

4.2. The Dynamics of Industrial Catalysts Under Operando Conditions

To showcase the complexity of an industrial catalyst and how to tackle it with recently emerged molecular modeling techniques, we illustrate the work of Bonati et al. (26). They studied the dynamics, adsorption, and bond breaking of the N_2 molecule when interacting with the (111) iron surface, one of the reactions taking place during the Haber-Bosch process. Interestingly, they found that the surface is very dynamic under operating conditions and that active sites are only formed thanks to the extreme operating conditions. Figure 3a–c gives a summary of their main results at two different temperatures. The plots show the resulting FES in terms of the N–N distance that needs to break during the reaction and the N_2 partial charge. These charges give an indication of the critical charge that needs to be transferred to sufficiently weaken the bond before it is broken. The FEPs are shown at a low temperature and high temperature, the latter of which is representative for the true operating conditions.

At low temperatures, one clearly finds metastable states characterized by distinct N_2 charges. The transferred charge varies from values close to zero when not interacting with the surface

FES: free energy surface

RI-RPA: resolution of identity random phase approximation

to systematically higher values. At higher temperatures, the metastable states are less clearly defined and do not correspond to classical adsorption states. Instead, the metastable states now more closely correspond to an ensemble of mostly disordered defected structures. Some precursor adsorption states for the reaction (α') are no longer metastable. The low-temperature prereactive state corresponds to a pocket where nitrogen can make a sevenfold coordination with iron in order to optimize the charge transfer. At high temperatures, such well-defined cavities do not exist anymore and cavities are continuously formed and broken. N_2 must find a critical configuration to weaken its bond. The example of the breakage of the nitrogen bond on iron surfaces clearly shows that the reactive behavior at high temperatures cannot be extrapolated from the observations at low temperatures.

The work summarized above was possible thanks to the derivation of MLPs and enhanced sampling MD simulations to follow the adsorption and reaction on the fly at operating conditions for sufficiently long timescales. To obtain more insight into the chemistry of this process, Bonati et al. (26) additionally trained a second ML model to estimate the partial charges of nitrogen without having to perform systematically expensive QM calculations. The approach is inspiring, demonstrating how various properties can be learned in a layered way.

Some of the effects observed in this case study are not unique to metal surface catalysis, as they have also been observed in other areas of catalysis. The existence of broad transition state regions and occurrence of a multitude of transition states were also observed in catalytic cracking taking place at very high temperatures ($>500^\circ\text{C}$) (108). For methylation reactions in the context of the MTH process taking place in large-pore zeolites, reacting molecules could easily rotate and translate, and it was also not possible to pinpoint one transition state characteristic for the chemistry at operating conditions (110). In the last decade, first-principles MD simulations have become more popular for revealing insights at operating conditions. However, as these simulations are very expensive, the community is shifting more toward performing MD simulations using MLPs. The key challenge lies in training accurate reactive MLPs that are able to capture the complex chemistry. Once such MLPs are available, they provide a perspective to reveal the nature of chemical processes under realistic operating conditions in a much more efficient way.

5. THE COMPLEXITY OF INDUSTRIAL CATALYSTS AND REACTIVE PATH DISCOVERY

A great challenge in modeling industrial catalysts at relevant operating conditions is the discovery of dominant reactive paths. Evidently, detailed knowledge of the reaction network is critical to further develop multiscale modeling methodologies like microkinetic modeling or perform kinetic MC schemes (122–124). The dominant chemistry for a given chemical transformation will depend on both changes in reaction conditions and the dynamics of the (components of the) heterogeneous catalyst. An illustrative example is the conversion of CO_2 and syngas (a CO/H_2 mixture) to valuable C_2 – C_4 olefins or valuable aromatics such as benzene, toluene, or xylene using heterogeneous bifunctional catalysts. In this two-step process, reactants are first activated over metallic active sites in bulk oxides to a wide range of C_1 intermediates. In the second step, the complex product distribution coming from the bulk oxide can be upgraded over zeolites (125). In this process, fundamentally different types of materials and active sites must cooperate to produce the target molecules. **Figure 4a** shows a proposed oxygenate-mediated catalytic cycle for the conversion of syngas to olefins and aromatics, showing how the cooperation between the bulk oxide and zeolite can lead to complex reaction cycles (126).

A long-established method in catalyst design is a bottom-up approach to compute rate constants from first-principles calculations with idealized structural models of the heterogeneous

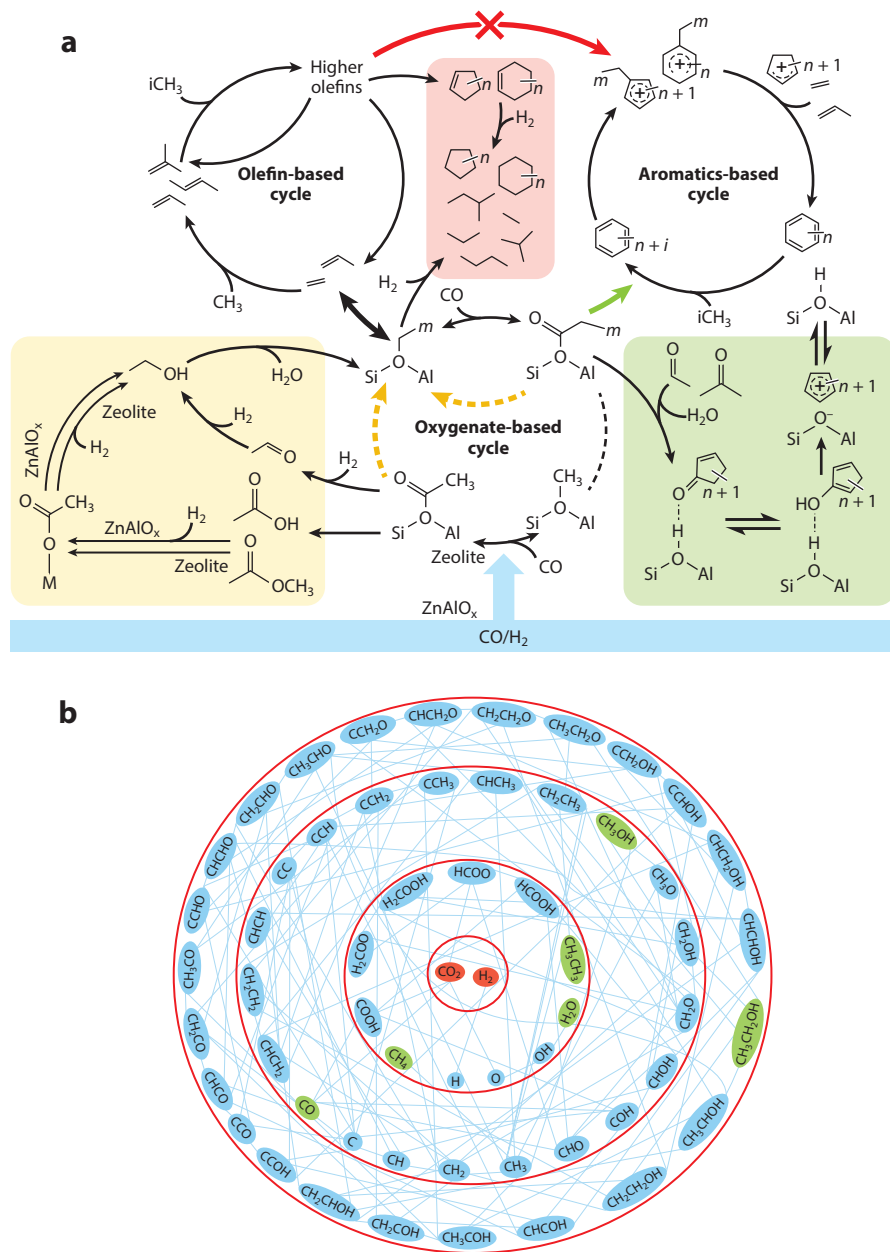


Figure 4

(a) Proposed mechanism of syngas (CO/H_2 mixture) conversion on $\text{ZnAlO}_x/\text{H-ZSM-5}$. Adapted with permission from Reference 126. (b) Example of using graph theory to represent a complex reaction network for CO_2 hydrogenation on Pd_2Cu . Adapted with permission from Reference 127; copyright 2024 American Chemical Society.

catalyst to identify the rate-determining steps in a (often simplified) reaction network. Mapping complex reaction networks to a graph, as shown for catalytic CO_2 hydrogenation on a heterogeneous catalyst in **Figure 4b** (127), not only enables systematic exploration of competing reaction mechanisms but also enables automatic reaction network exploration (127–129). However, the

necessary geometry optimizations and transition state searches to compute rate constants for all possible elementary steps of a complex reaction network necessarily result in a substantial (and often intractable) computational overhead.

An attractive alternative is a top-down approach guided by kinetic measurements to construct kinetic phase diagrams and identify reactivity descriptors, such as adsorption energies of intermediates. This second approach is necessarily limited by the lack of sensitivity of all possible elementary steps of the reaction network as well as the limited spatial and temporal resolution of experiments to the experimental observables (130, 131).

The recent advent of ML shows great promise for accelerating catalyst design in both approaches. First, the development of accurate MLPs enables high-throughput screening of reaction networks in the bottom-up approach by substantially lowering the computational overhead associated with energetic evaluations. A second major application of artificial intelligence in catalyst design is the unparalleled ability for pattern recognition in large and complex datasets, which is applicable in both reaction network exploration (bottom-up approach) and descriptor identification (top-down approach). Automatic workflows for heterogeneous catalysis are actively being developed, bridging timescales and length scales by combining automated reaction network exploration with reactor-scale computational fluid dynamics (124). While other multiscale modeling approaches to zeolite catalysis are available in the literature, applications remain scarce due to the inherent complexity of chemical transformations in nanoporous materials, as highlighted throughout this review (132).

6. A TIME PERSPECTIVE OF THE VARIOUS STEPS TAKING PLACE DURING A CATALYTIC OR SEPARATION EXPERIMENT

Within this section more insight is given into the various steps taking place during a realistic catalytic or separation experiment when looking at it from the crystal particle. Additional spatial heterogeneities resulting from encapsulating the material in binders or larger macroscopic bodies are not considered. During a catalytic or separation experiment, a feed of molecules is sent over a nanoporous material and a series of events takes place, as illustrated in **Figure 5a**. First, the molecules enter the crystal, possibly after interacting with external active sites; then they diffuse through the lattice to find interesting places—the active sites—where they can adsorb/react. The intracrystalline diffusion may be hindered, for example, due to the presence of narrow windows in the material or due to the presence of other bulky molecules. After a successful reaction, the formed products desorb from the active site and diffuse toward the exit to finally leave the material. Timescales for all events vary in a very broad range from picoseconds for molecular vibrations to seconds/hours for activated events (19, 20). Ideally, one would obtain a consistent dynamics view of all these processes from the molecule to the crystal particle level. However, this poses enormous challenges and is not yet possible with current modeling approaches. Reactions are local events taking place on the (sub)nanometer level and are typically described by QM methods. Diffusion, on the other hand, is a nonlocal event taking place at the 10–100-nm scale, and because of this, it is usually described by means of classical FFs, which usually implies that specific host-guest interactions are not accurately accounted for. Obtaining an overall time perspective of the whole trajectory is complicated by the vastly different length scales and timescales involved. Furthermore, such a consistent dynamics view would require a consistent level of theory, with QM accuracy to describe the interatomic interactions for all events. In the following, insight is given on how to model the various events. It will become clear that further methodological advancements are necessary to capture the intricate interplay between reaction and diffusion, along with other challenges critical for achieving a comprehensive, time-resolved understanding of molecular trajectories through the lattice.

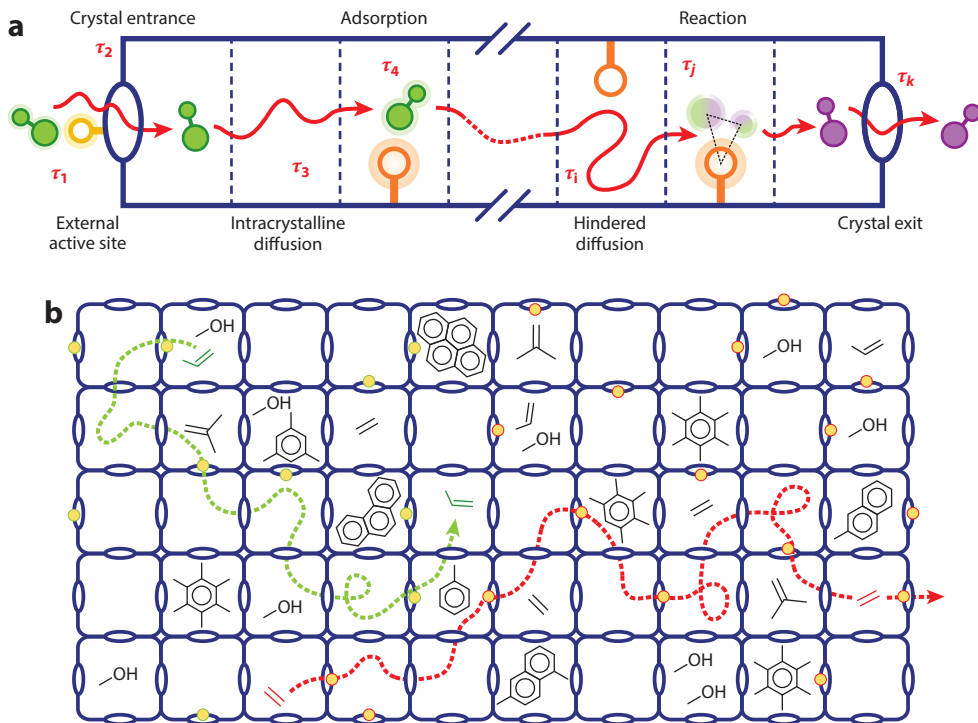


Figure 5

(a) Various steps taking place during a typical catalytic experiment. Molecules enter the crystal and diffuse through the pores to arrive at the active site to adsorb/react. Diffusion in/out of the crystal can be hindered due to external active sites, other molecules, or narrow windows in the crystal. (b) Visualization of methanol-to-hydrocarbons conversion in an industrial catalyst. As the process evolves, bulkier hydrocarbons are formed and confined in the nanopores, leading to additional barriers for diffusion for smaller molecules (green and red paths).

As outlined in Section 3, we have witnessed a major evolution in how to determine reaction kinetics of activated processes. This evolution ranged from (a) utilizing a static approach to TST using the harmonic approximation to (b) using dynamically sampled FESs in terms of well-chosen CVs and to (c) finally going beyond TST using, e.g., TPS/TIS or the Bennett-Chandler theory (81).

These methods provide information on the intrinsic kinetics of catalytic reactions, referring to the state where all reacting species are adsorbed. To connect with experiments, one requires apparent kinetics, where the thermodynamics of the adsorbed state is also taken into account. The common approach to include adsorption thermodynamics currently mostly relies on a static approximation; however, to obtain consistent estimates of apparent rates comparable to experiment, dynamic sampling (MD or MC) of adsorption at the QM level is necessary. Fortunately, the boost in efficiency provided by MLPs has recently allowed computation of adsorption enthalpies using GCMC calculations for CO_2 in ZIF-8 and Mg-MOF-74 at DFT accuracy (114). Similarly, adsorption and diffusion of H_2 were studied in Al-soc-MOF-1d using GCMC and MD at DFT accuracy (133). For the case of adsorption of isobutene in H-SSZ-13, free energies of adsorption were obtained using accurate MLPs, which also allowed disentanglement of the separate enthalpic and entropic contributions (86). Various complementary ML methodologies have been proposed to determine with near chemical accuracy adsorption enthalpies for industrially

relevant processes (67). Despite these advances, we are not aware of apparent rates of catalytic reactions in nanoporous materials determined from a consistent dynamic description of both the adsorbed state and intrinsic reaction rates.

Diffusion of reacting species and formed products is an essential component of any catalytic cycle. Also, in separation experiments, transport phenomena are important as the separation can be thermodynamically driven but also kinetically driven, i.e., when one component diffuses faster through a membrane than another (134). Diffusion may also become hindered, e.g., when molecules have to pass through narrow windows, effectively turning diffusion into a rare event. As a result, the traditional methodology of extracting self-diffusivities using Einstein's mean square displacement (MSD) will not be efficient. As a result, more advanced methodologies are required, such as a hopping model where molecules hop from one cage to another. Using this hopping model, FEPs of diffusion can be constructed, from which diffusivities are extracted using TST (135–137). Lastly, unlike commonly assumed, diffusion and reaction can be intertwined, as illustrated below for a practical example.

Previous concepts are illustrated for a realistic industrial MTH catalyst in **Figure 5b**. Ethene and propene are essential MTH products and need to diffuse through the crystal. With time on stream, the catalyst evolves and bulkier hydrocarbons are formed and confined within the pores of the zeolites, leading to additional barriers for the transport of smaller species (58, 138). Interestingly, it was found that ethene or propene interacts with Brønsted acidic protons of the framework, which facilitates diffusion (135, 139). This effect was not properly captured by classical FFs, and QM-based methods accounting for specific interactions were necessary. Diffusion can also be highly dependent on the operating conditions; as such, it was found that higher propene loadings enhanced diffusivity (135, 139). During the MTH process, ketene is also present as an important precursor to form C=C double bonds. It was recently discovered that ketene can react to other species when diffusing through the eight-membered rings of H-SAPO-34 in the presence of water (140). This example clearly illustrates that for an industrial catalyst, reaction and diffusion can be entangled and that diffusion strongly depends on the operating conditions. For unhindered diffusion, self-diffusion coefficients can in principle also be determined from the MSD of adsorbate molecules from very long MD runs (141, 142). Given the excessive long simulation times, this has mostly been done using classical FFs; we see future avenues to determine these transport properties with QM accuracy based on accurate MLPs. Future modeling efforts should be directed to modeling reaction and diffusion in a consistent way and to building kinetics models encompassing both reaction and diffusion at operating conditions.

Another event of the molecular trajectory involves the entry of molecules from the external surface into the crystal. Experimentally, there are clear indications for surface barriers, induced by pore blocking and/or amorphous outer layers, even though the precise molecular origin of these barriers is poorly understood (134). To address this, several computational approaches have been proposed, including nonequilibrium and equilibrium MD, to quantify surface resistance through calculation of molar fluxes or diffusivities across external surfaces (143–148). While these methods offer valuable insights, their application to realistic materials remains challenging. Accurate simulations require large-scale systems and long timescales, particularly when attempting to capture the complexity of surface interactions and structural heterogeneities. Various simulations have been done using FFs to understand how diffusion is affected by spatial heterogeneities such as grain boundaries and external surfaces (146). An interesting case is diffusion of benzene through silicalite (146, 149). Starting from realistically terminated surfaces and a reservoir of benzene molecules, various diffusivities could be derived. Future endeavors in this field should start from realistic models for the external surface and well-trained MLPs to represent the interaction with entering guest molecules that allow dynamic sampling on longer length scales and timescales. To

proceed further in this area, good benchmark systems need to be established experimentally with well-characterized external surfaces.

SUMMARY POINTS

1. Nanoporous materials represent a broad and highly versatile class of materials—including zeolites, zeotypes, metal-organic frameworks (MOFs), covalent organic frameworks, mesoporous silicas, and porous carbons—which can have different functions, such as catalysis, separation, and capture/storage, that are key for future technologies.
2. A typical molecular modeling exercise consists of four main components: a realistic representation of the molecular structure, a method to evaluate the energies and forces to construct the highly dimensional potential energy surface (PES), algorithms to sample the PES under relevant thermodynamic conditions, and methods to extract the properties of interest.
3. Industrial catalysts often operate under extreme conditions of temperature and pressure and need to handle complex feedstocks. To unravel the underlying chemistry at these operating conditions—including the nature of reactive intermediates, reactive pathways, and active sites—it is essential to account for the dynamic behavior at the relevant industrial conditions.
4. The function of industrial catalysts is governed by very complex reaction cycles, where many reactions take place simultaneously, making the discovery of reactive paths at operating conditions extremely challenging.
5. Industrial catalysts evolve with time on stream, where their structure changes and the reactive paths may change, which is still very difficult to simulate with current modeling strategies.
6. During a typical catalytic or separation experiment, where guest molecules are sent over a nanoporous material, a sequence of events occurs, including diffusion, adsorption, and reaction. These events span a wide range of length scales and timescales, making it very challenging to obtain a unified dynamic view of the entire molecular trajectory.
7. The integration of machine learning methods in the field of molecular modeling significantly advanced the study of chemical transformations in nanoporous materials and industrial catalysts, providing methods for the derivation of machine learning potentials (MLPs) to describe in a more efficient way the PES toward pattern recognition and graph exploration to unravel reaction mechanisms.

FUTURE ISSUES

1. Methodological advances are needed to build structural models for realistic crystal particles used in catalysis and separation, having mesoscale dimensions with local defects, correlated nanoregions, mesoporosity, internal and external surfaces, and their interaction with bigger units of catalyst pellets.
2. Synergistic experimental modeling endeavors are necessary to unravel the structure of realistic nanoporous materials from the nano- to the mesoscale under operating

conditions with atomic precision. This includes setting up well-defined systems to be characterized with the most advanced imaging and spectroscopic techniques at the conditions where the material does the work.

3. New innovative methods are needed to capture the complex chemistry of industrial catalysts within the training data for MLPs and discover reactive paths in an automated way.
4. Further developments at the quantum mechanical level are necessary to treat systems with challenging electronic structures, such metal-loaded zeolites with open shell structures or MOFs with partially filled d- or f-shells. These advances are necessary to generate training data for the development of chemically accurate MLPs
5. Advanced path-based sampling methods and kinetic theories need to be developed, making optimal use of MLPs and allowing evaluation of the kinetics in a fully dynamic way beyond transition state theory.
6. Methodological advances are necessary to build consistent kinetic multiscale models for diffusion, reaction, and adsorption, with disparate length scales and timescales, in order to obtain an overall dynamic view for local and nonlocal events and from the molecular to the crystal particle level.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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